CONTINUED ON DRAWING 25\E\1065-001 2000-F25-SLD-EL-10010

3F 1 — 415V 3 PHASE 50Hz 3150A FORM 3 FAULT LEVEL 50kA FOR 1 sec.

4F 1 | 4F 2 | 4F 3 | 4F 4 | 4F 5

5F 1 | 5F 2 | 5F 3 | 5F 4 | 5F 5 — 415V 3 PHASE 50Hz 200A FORM 3 EMERGENCY SUPPLY BUSBAR FAULT LEVEL 18kA FOR 1 sec.

3000/5

0-500V

0-3000A

4-20mA TO PROCESS CONTROL CUBICLE

2A/20A

415/110V

2A

VAMP 221 ARC FLASH RELAY — NOTE 1

INTEGRAL C.B. 3150A

INST

NEUTRAL

EARTH

200/5

ROGOWSKI COILS

200A/200A | 200A/200A | 250A/250A | 63A/63A

CA7-43

CEP7 16-25

200A — 200A

16A/32A

415/110V

A N

L1(E) N1(E)

25A/32A | 25A/32A | 16A/32A

BYPASS TRANSFORMER | BYPASS TRANSFORMER

25-XE-62 | 25-XE-63

11 kW

25-DB-62 | DB No.1 | DB No.2

25-CP-83
MAIN C/B & RECLOSER REMOTE ISOLATION
DWG. 25\E\1086-019

25-TR-31
33kV/415V
2000 kVA
Dyn11

HV EARTH

SUBSTATION MV EARTH BAR

EARTH GRID

TO SOUTH SWITCHYARD (D2) #8 - 3H RECLOSER REF. DWG. 180\E\3045

BUS TIE INTERLOCK FROM BUS TIE 3250A ISOLATOR REF. DWG. 25\E\1065-001

33kV 3Ph SUPPLY FROM SOUTH SWITCHYARD (D2) #8 - 3H RECLOSER REF. DWG. 180\E\3045

DIST. BOARD 36A
415V NORMAL DB 36A
DISTRIBUTION BOARD
REF. DWG. 25\E\1068-001

FEED TO FORM 1 & EMERGENCY SECTIONS J & H

25-CM-31B
FLOTATION AREA 2
COMPRESSOR

30-PS-33
FLOTATION TAILS WASHDOWN PUMP
REF. DWG. 25\E\1074-004

SPACE

415V EMERGENCY SUPPLY
FROM SUB STATION No.7 80-SW-01
CB 25-SW-31 GENSET 80-GE-61
REF. DWG. 80\E\1132

110V CONTROL TRANSFORMER
REF. DWG. 25\E\1074-001

TO 25-DB-62
REF. DWG.

25-XE-62
BYPASS TRANSFORMER - UPS 1
REF. DWG. 25\E\1070-003

TO DB No.1
REF. DWG. 25\E\1070-001

25-XE-63
BYPASS TRANSFORMER - UPS 2
REF. DWG. 25\E\1070-002

TO DB No.2
REF. DWG.

25-PC-01
SUB 6 FEEDER B

415V FEED FROM TEIR 5F

12

CONTINUED ON DRAWING 25\E\1065-001 2000-F25-SLD-EL-10010

4R 3 | 4R 4 | 4R 5 | 4R 6 | 4R 7 | 4R 8 | 4R 9 | 4R 10 | 4R 11 | 4R 12 — 415V 3 PHASE 50Hz 200A FORM 1 FAULT LEVEL 18kA FOR 1 sec.

25A

5R 1 | 5R 2 | 5R 3 | 5R 4 — 415V 3 PHASE 50Hz 200A FORM 3 EMERGENCY SUPPLY BUSBAR FAULT LEVEL 18kA FOR 1 sec.

100A/100A | 80A/80A | 80A/80A | 80A/80A

3TF46

3UA58 32-50

BYPASS SWITCH | BYPASS SWITCH

22 kW

SPARE | SPACE | SPACE | SPACE | SPACE | SPACE | SPACE | SPACE | SPACE | SPACE

45V FEED FROM TEIR 5F

DIST. BOARD 36B
415V EMERGENCY DB 36B
DISTRIBUTION BOARD
REF. DWG. 25\E\1069-001

30-TH-31
CONCENTRATE PRE-LEACH THICKENER
2000-F25-SCM-EL-10001

25-XE-64
CONTROL ROOM UPS
REF. DWG. 25\E\1090-001

25-DB-12
240VAC CONTROL ROOM
DISTRIBUTION BOARD
REF. DWG. 25\E\1070-004

25-XE-65
CONTROL ROOM UPS
REF. DWG. 25\E\1090-002

25-DB-13
240VAC CONTROL ROOM
DISTRIBUTION BOARD
REF. DWG. 25\E\1070-005

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Si [redacted] /2025
C

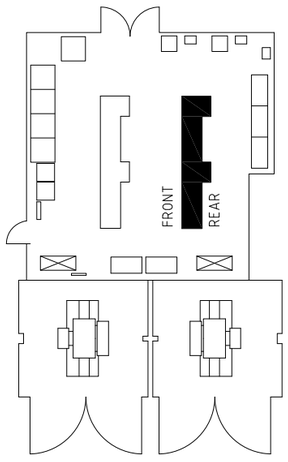
ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

LEGEND

LOCATION

SUB STATION 6

25-SW-31



NOTES

1. ALL AFS (ARC FLASH SENSORS) ARE CONNECTED TO THE ARC FLASH RELAY VIA I/O UNITS

LEGEND
- EXISTING
- NON-EXISTENT
- STAGE 1 - NEW
- STAGE 1 - REMOVE
- STAGE 1 - DECOMMISSIONED (OTHERS)
- STAGE 2 - NEW
- STAGE 2 - REMOVE
- STAGE 2 - DECOMMISSIONED (OTHERS)

KCGM GROWTH PROJECT

DRAWING NO. 2000-F25-SLD-EL-10011

REV	DATE	DESCRIPTION	CHECK	TECH APP
0	10/11/25	APPROVED FOR CONSTRUCTION	[redacted]	DM

MK	DATE	REVISION	BY	CKD	APP
12	27/10/21	RECLOSER PROJECT AS BUILT	RH	RH	MM
11	08/10/19	2019 S.L.D. UPDATE PROJECT "AS BUILT"		RH	
10	08/08/19	2019 S.L.D. UPDATE PROJECT "AS BUILT"- FIELD REVIEW		RH	
9	12/07/18	ARC FLASH SENSOR S INSTALLED		JMR	
8	21/10/14	25CM31B RELOCATED FROM 2F/1 TO 4F/3 DURING SHUTDOWN		SK	
7	22/05/14	AS BUILT		SK	

MK	DATE	REVISION	BY	CKD	APP
6	08/04/09	CONTROL ROOM UPGRADE	TW		
5	29/10/07	DCS UPGRADE ALTERATIONS			
4	09/11/04	AS BUILT 2004			
3	15/01/03	UPDATE EMERGENCY SUPPLY CHANGES			
2	03/02/01	MINOR CHANGES ADDED	SDES	AB	
1	01/10/00	ASBUILT MODIFICATION ADDED		KCGM	KCGM

REFERENCE DRAWINGS

DWG NO.

SCALE	NTS	
DRAWN	[redacted]	01/10/00
DESIGNED	KCGM	01/10/00
CHECKED	KCGM	01/10/00
APPROVED	KCGM	01/10/00
PROJ.APP	KCGM	01/10/00
PROJ. FILE #.		

FIMISTON PLANT
FLOTATION
SUBSTATION No.6 415V MCC 25SW31
SINGLE LINE DIAGRAM

SIZE	DRAWING NUMBER	REV
A1	25\E\1065-002	12

KALGOORLIE CONSOLIDATED GOLD MINES PTY. LTD. ABN 009 377 619 003